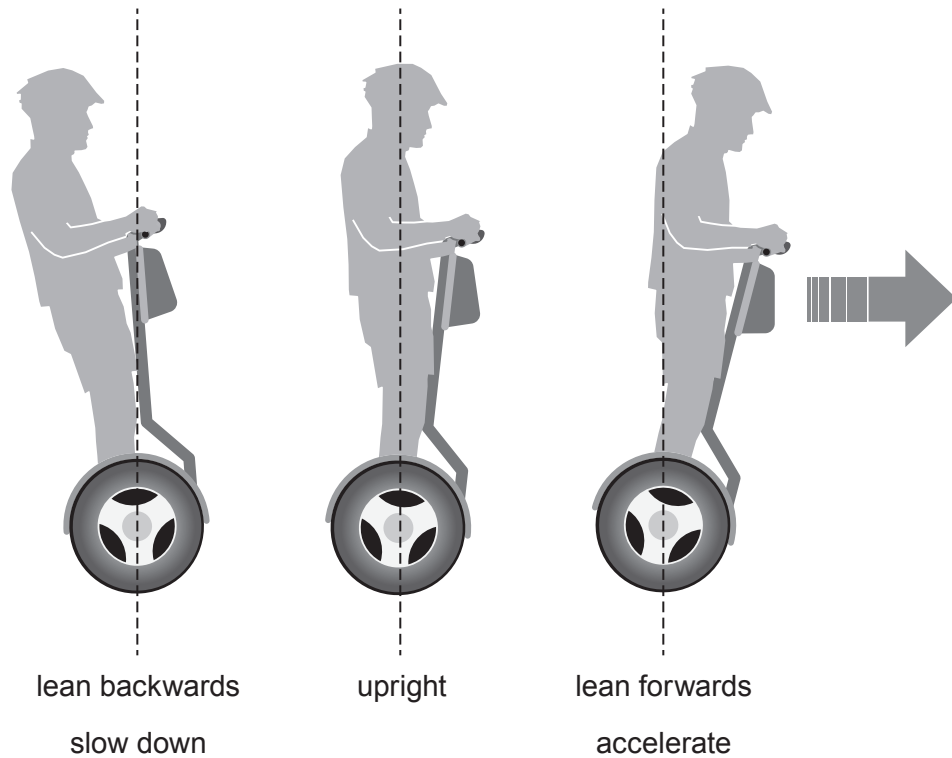
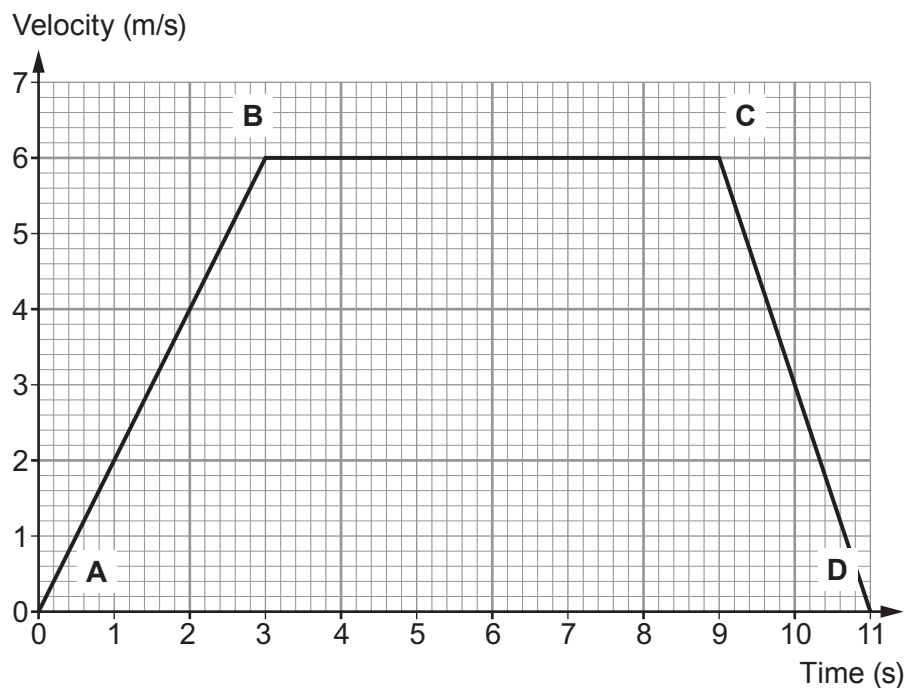


3. A Segway vehicle can be used as a method of transport for a person. The person leans forwards to make the Segway accelerate and leans backwards to slow it down. It is recommended that short training sessions are carried out before using the Segway on a busy pavement.



The velocity-time graph below relates to a short Segway training session that is carried out on a **dry** pavement. The person riding the Segway is instructed to carry out an emergency stop during the training.



Use information on the graph opposite to answer the following questions.

- (a) Between which points, **A to B**, **B to C** or **C to D**, is the person leaning forward on the Segway? [1]

- (b) Use the equation:

$$\text{acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

to calculate the acceleration during the first 3 seconds and state its unit. [3]

Acceleration =

Unit =

- (c) The combined mass of the person and the Segway is 110 kg.
Use the equation:

$$\text{resultant force} = \text{mass} \times \text{acceleration}$$

to calculate the resultant force during the first 3 seconds. [2]

Resultant force = N



- (d) The reaction time of the person riding the Segway is 0.6 seconds.
Use the graph on page 8 to determine the time, during the training session, at which the rider is asked to carry out the emergency stop. [2]

Time = s

- (e) Newton's first law of motion states "An object will remain at rest, or move at constant velocity, if the forces acting on it are balanced".

A student states that during the Segway training session balanced forces act on it for 11.0 s. Explain whether you agree with the student. [2]

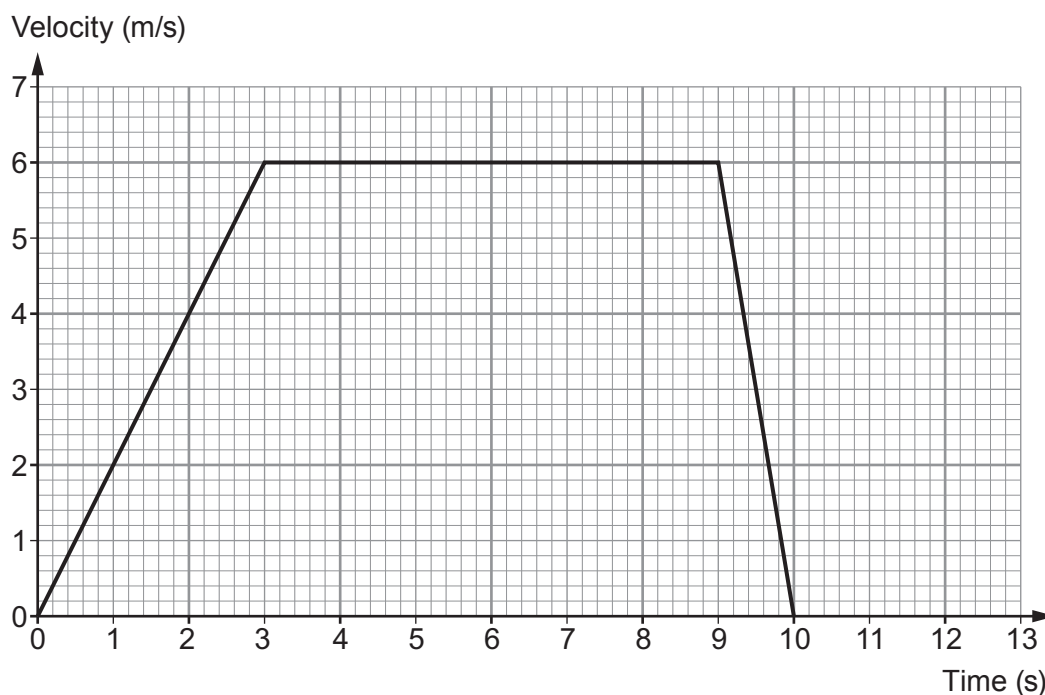
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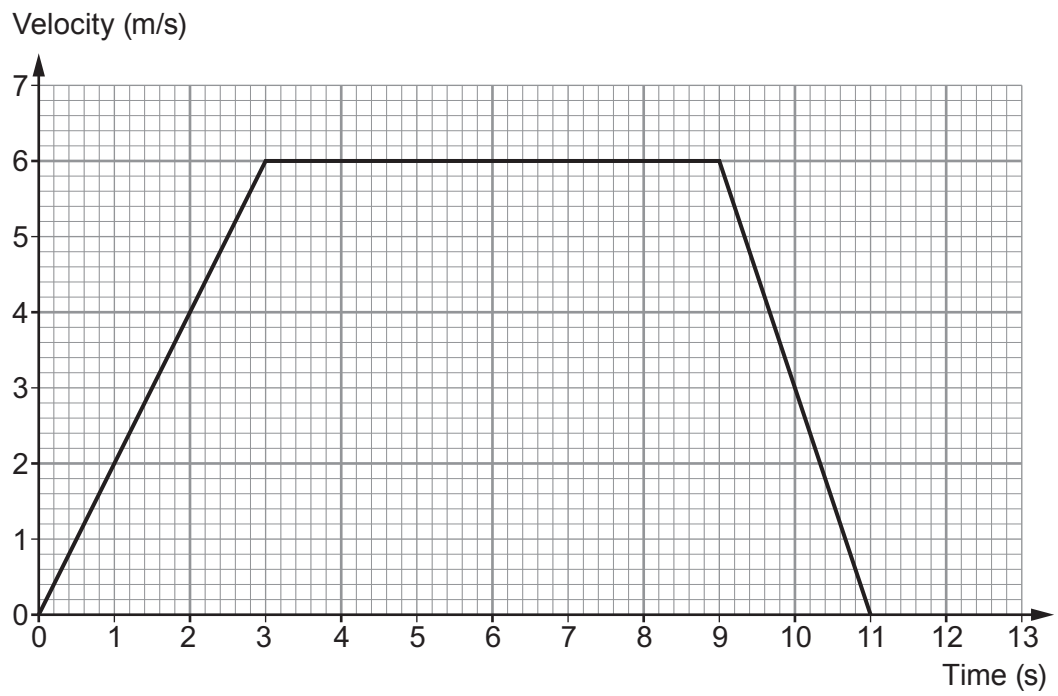
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- (f) The graphs below and opposite show 3 possible training sessions carried out by the same Segway rider. Lucy says that **Graph C** would represent the emergency stop carried out on an **icy** pavement. Explain whether you agree with Lucy. [2]

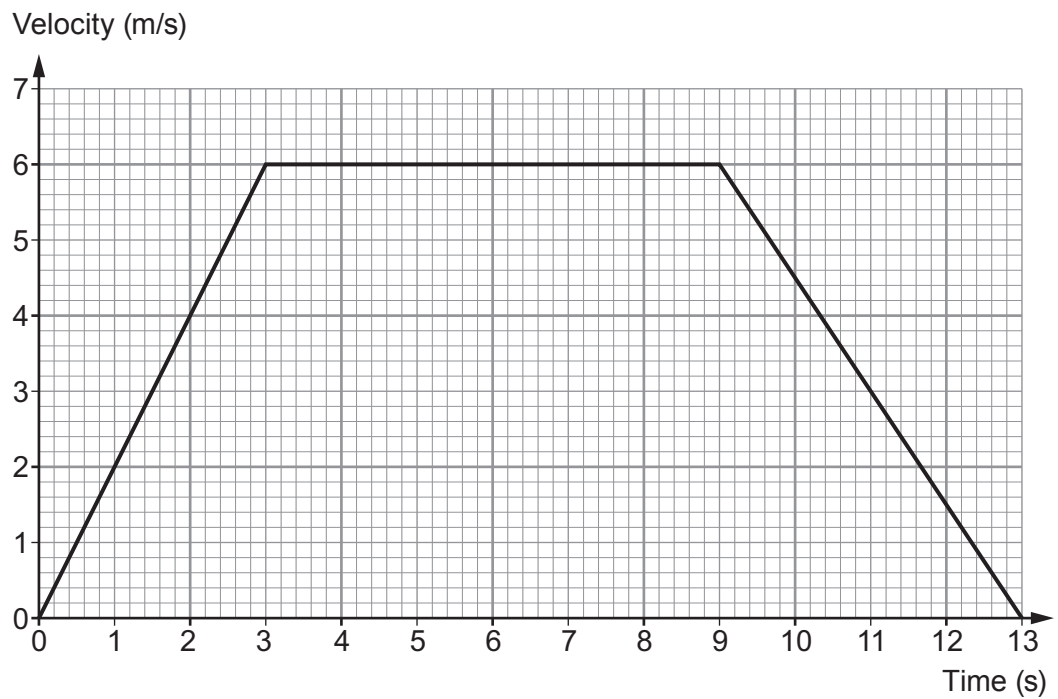
Graph A



Graph B



Graph C



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